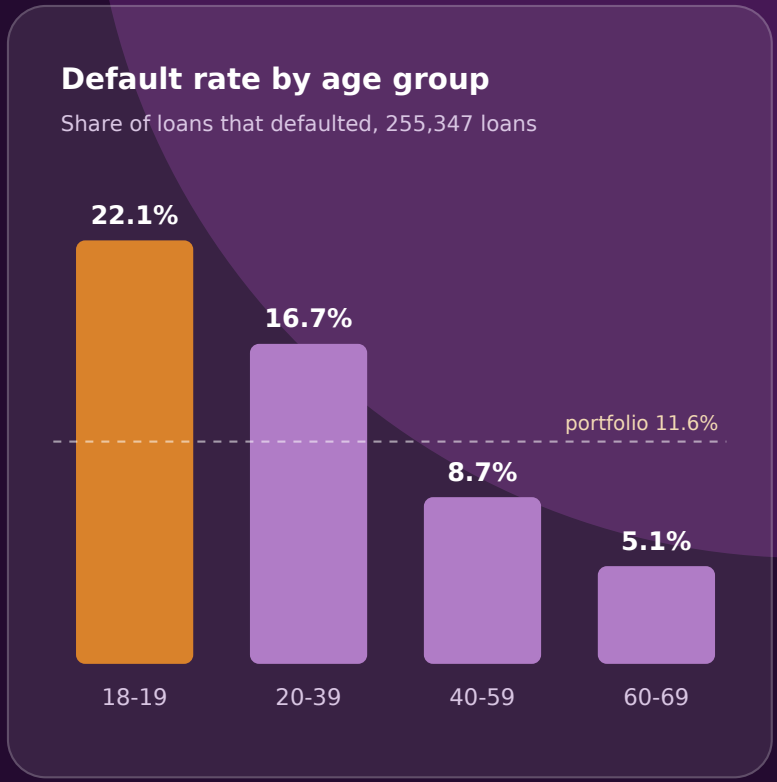
BUSINESS INTELLIGENCE PROJECT

Loan Default Analysis

Where default risk sits in a lending portfolio, and what a lender can do about it



Power BI | DAX | SQL Server | Dataflows

255,347
loans analysed

11.6%
portfolio default rate

14
dashboard visuals

2013-2018
period covered

Project Documentation | October 2026

SQL Server > Power BI Dataflow > Power BI model > 3-page dashboard

Executive Summary

This project analyses 255,347 loans issued between 2013 and 2018 to show where default risk sits in a lending portfolio. The data is stored in SQL Server, published through a Power BI Dataflow, modelled in Power BI with DAX, and presented in a three-page dashboard: **Loan Default and Overview**, **Applicant Demographics and Financial Profile**, and **Financial Risk Metrics**.

Key findings

- About 1 in 9 loans defaults (11.6%), and the yearly rate has stayed between 11.5% and 11.7% for six years.
- Risk depends on who borrows, not when: default rate falls from 22.1% (ages 18 to 19) to 5.1% (ages 60 to 69), and from 22.0% (income up to 30K) to 9.5% (income above 60K).
- Interest rate and job tenure are also strong signals: default rises from 6.0% to 17.8% across interest rate bands and falls from 16.2% to 7.6% as months employed increase.
- Loan size is almost the same in every group (about 127K), so the highest-risk groups receive the same loan as the lowest-risk groups.
- By amount, 13.2% of the money lent is in defaulted loans, higher than the 11.6% by count.

Project highlights

- **End-to-end pipeline:** from a SQL Server table to a published Power BI report, with a Dataflow layer between them.
- **Business-ready model:** four calculated columns turn raw fields into age groups, credit score bins, income brackets and year; thirteen measures drive 14 visuals.
- **Validated analysis:** default rates were checked across 14 borrower and loan characteristics, and every breakdown reconciles to the same 255,347 loans and 29,653 defaults.
- **Accurate measures:** each chart was checked against the data and the DAX refined where needed, so every chart measures what its title says.
- **Clear evidence trail:** verified facts, derived figures and recommendations are kept separate.

1. Business Problem and Objectives

A loan default is a loan the borrower fails to repay as agreed. For a lender it is the main source of credit loss, and it is not spread evenly across customers. A lender therefore needs to know how many loans default, where, and among whom, and whether the pattern is changing.

Objectives. Monitor the portfolio default rate and its trend; profile borrowers by demographics and finances; track year-over-year change in lending and defaults; identify the segments where default is concentrated; and lay a foundation for deeper statistical work.

Stakeholder	How they would use the dashboard
Credit risk and portfolio managers	Track default rate and exposure by purpose, credit tier and income; spot concentration.
Underwriting and policy	Decide which applicant characteristics merit closer verification or different limits.
Finance and leadership	See lending growth and default trends on one page and decide where to investigate.

Questions the analysis answers

- What share of loans default, and is that share rising or falling?
- Which loan purposes, employment types, age groups and income levels carry the most default risk?
- How much lending, by amount, is exposed in each segment?
- Do borrower characteristics such as income, tenure, interest rate and credit score separate safe loans from risky ones?
- Are defaults growing faster or slower than lending volume?

Decisions the analysis can support

- **Lending decisions:** identify segments where tighter verification, lower limits or different pricing should be tested.
- **Portfolio monitoring:** a stable baseline default rate (11.6%) with alert bands, and segment rates tracked alongside it.
- **Risk management:** concentration checks by purpose and credit band, and early warning when defaults grow faster than lending volume.

2. Dataset and Data Preparation

The **Loan_default** table holds one row per loan. All LoanIDs are unique, and the ranges below were verified directly in the model.

Group	Columns (verified range or values)
Applicant	Age (18 to 69), Education (High School, Bachelor's, Master's, PhD), MaritalStatus (Single, Married, Divorced), HasDependents
Employment and income	EmploymentType (Full-time, Part-time, Self-employed, Unemployed), Income (15,000 to 149,999), MonthsEmployed (0 to 119)
Credit profile	CreditScore (300 to 849), NumCreditLines (1 to 4), DTIRatio (about 0.1 to 0.89), HasMortgage
Loan	LoanAmount (5,000 to 249,999), LoanPurpose (Home, Business, Education, Auto, Other), InterestRate (below 5% to above 25%), LoanTerm (12 to 60 months), HasCoSigner, Loan date (2013 to 2018)
Outcome	Default (True/False): the basis of every default rate

The data at a glance

255,347 loans, every LoanID unique	**2013-2018** loan dates, 1 Jan 2013 to 31 Dec 2018	**29,653** defaulted loans (11.6%)	**32.58bn** total loan amount	**127.6K** average loan (derived)

How the columns are used. Default is the outcome behind every default rate and LoanAmount is behind every amount. The remaining columns are used to segment borrowers, and four of them (Year, Age groups, Credit score bins and Income Bracket) are derived in the model, as described in Section 2.2.

2.2 Data Preparation and Quality

Data preparation. The Dataflow reads the SQL table as it is, and the Power BI query only navigates to the Dataflow entity, so data types come from SQL. The analytical preparation is done in four DAX calculated columns that create business-friendly segments:

Column	Rule	Purpose
Year	Year of the loan date	Yearly trends, year-over-year change
Age groups	Teen: up to 19; Adults: 20 to 39; Middle aged: 40 to 59; Senior Citizens: 60 and over	Life-stage segments
Credit score bins	Very Low: up to 400; Low: 401 to 450; Medium: 451 to 650; High: above 650	Credit tiers
Income Bracket	Low: up to 30,000; Medium: 30,001 to 60,000; High: above 60,000	Affordability tiers

Data quality assessment

The checks below were run directly on the model with DAX queries.

Data quality check	Result
Duplicates	255,347 distinct LoanIDs for 255,347 rows; no repeated IDs.
Missing values and outliers	Zero blanks in LoanID, Age, Income, LoanAmount, CreditScore, EmploymentType, Default and loan date; every other analysed field produced group counts summing to exactly 255,347. All values fall within plausible, round bounds.
Group balance and bins	Employment types hold about 25% of loans each, purposes about 20% each, and loan sizes are spread evenly. This makes comparisons clean, but also suggests a public or simulated dataset, so its source should be recorded. Credit and income bins differ in width, so totals by bin follow the number of loans in each; rates are the better comparison.

3. Technical Architecture and Workflow

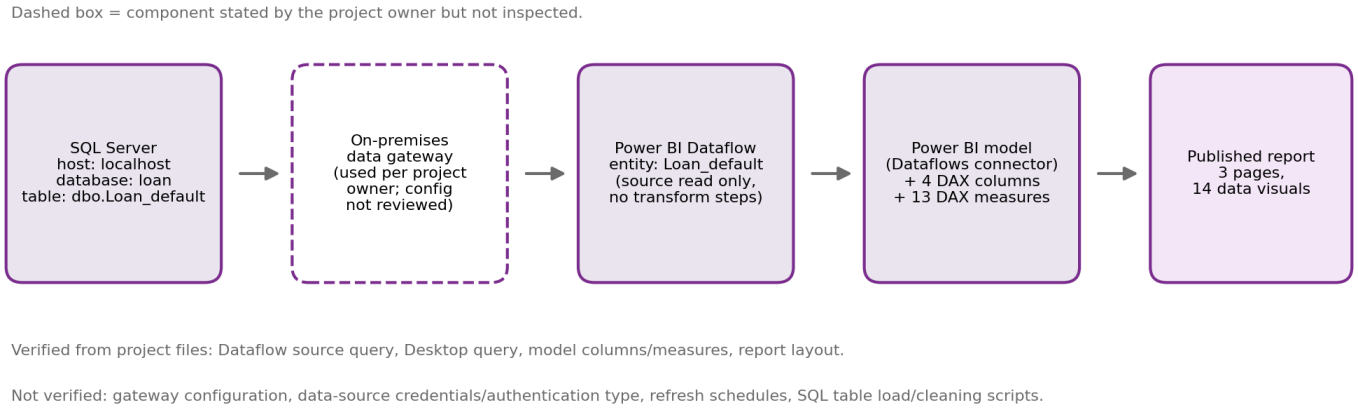


Figure 1. Workflow. Solid boxes are verified from the project files; the dashed box is stated by the project owner and was not inspected.

Component	Role in the project
SQL Server	Source system: database `loan` on `localhost`, table `dbo.Loan_default`.
On-premises data gateway	Connects the cloud Dataflow to the local SQL Server. A server named localhost cannot be reached directly from the cloud service.
Power BI Dataflow	Reads the SQL table and publishes it as the entity `Loan_default`, with no transformation steps.
Power BI model	Loads the Dataflow entity (the data is stored in the model), then adds the four calculated columns and thirteen measures.
Power BI report	Three pages and 14 visuals, published to the Power BI service.

Refresh and access. The design is two-stage: the Dataflow refreshes from SQL, then the dataset refreshes from the Dataflow. Refresh depends on the SQL Server being available, the gateway host being online and the user having access to the Dataflow workspace. Gateway settings, credential type and refresh schedules were not part of this review and should be added once documented.

4. Dashboard Walkthrough

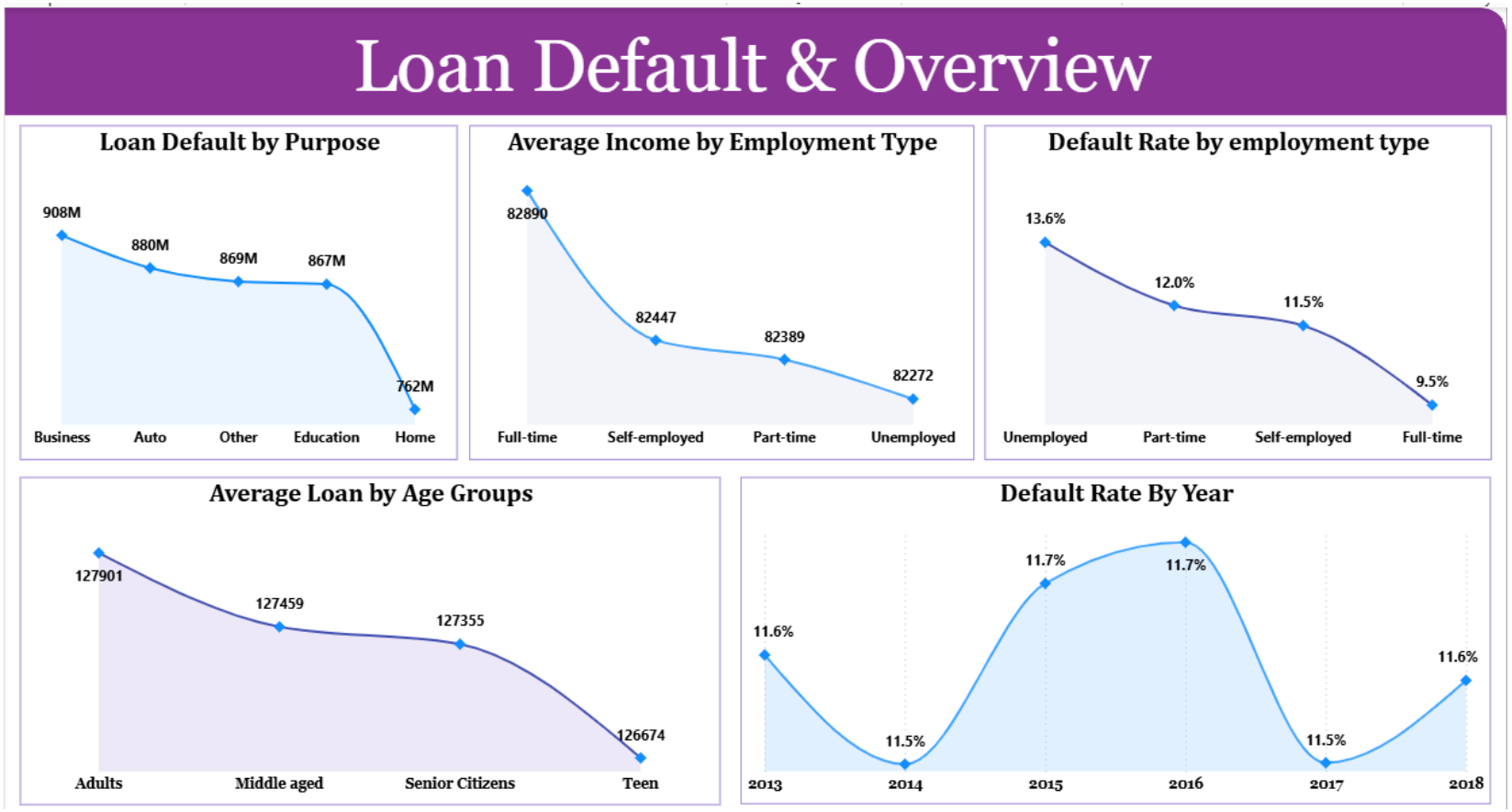
Concept	Meaning
Default rate	Defaulted loans divided by all loans in the group shown, so groups of different size compare fairly.
Mean vs median	The mean is the average; the median is the middle value and is less affected by extreme values.
Credit score bins	Scores grouped into Very Low, Low, Medium and High bands (definitions in Section 2).
Year-over-year (YoY)	(This year - last year) / last year, as a percentage.
Year-to-date (YTD)	A running total from 1 January to the latest date in view. The data ends on 31 December 2018, so the YTD visual shows the 2018 total.

How the dashboard is organised

- **Page 1, what is happening:** default by purpose, employment type and year, with average income and loan size for context.
- **Page 2, who is borrowing:** loan size and exposure by credit tier, age, marital status, household profile and education.
- **Page 3, how risk and volume move:** year-over-year change in lending and defaults, latest-year exposure, and an interactive drill-down by income and employment.
- **Built for reuse:** age groups, credit bins and income brackets are created once as calculated columns and reused across visuals, and the thirteen measures are grouped in three tables that mirror the three pages.

Dashboard Page 1: Loan Default and Overview

This page answers the question: what is happening in the portfolio?



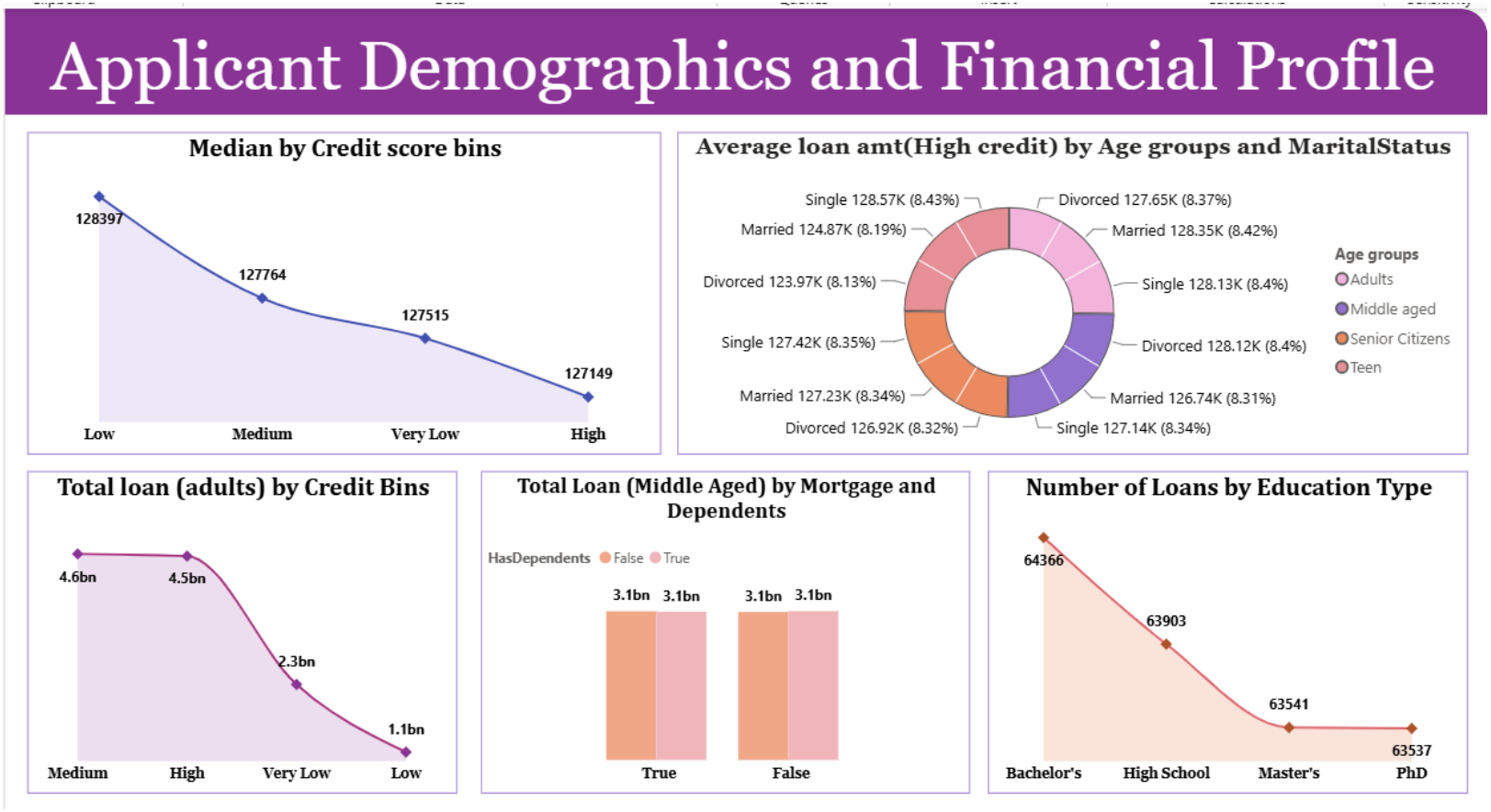
Page 1: Visual Details

#	Visual	What it shows and how it is calculated	How to read it and use it
1	Loan Default by Purpose	Sum of LoanAmount where Default = True, per purpose: Business 908M, Auto 880M, Other 869M, Education 867M, Home 762M (4.29bn in total).	Shows where lending is lost. Business is highest. Home is lowest: 11.6% of its lent amount defaulted, against 13.3% to 13.9% for the other purposes.
2	Average Income by Employment Type	Average Income per type: Full-time 82,890; Self-employed 82,447; Part-time 82,389; Unemployed 82,272.	The four types earn almost the same (within 1%). The axis is zoomed, so the decline looks steeper than it is.
3	Default Rate by Employment Type	Defaulted loans divided by all loans in each type: Unemployed 13.6%, Part-time 12.0%, Self-employed 11.5%, Full-time 9.5% (about 64,000 loans each).	Unemployed borrowers default about 4 points more often than full-time borrowers, so employment type is a candidate for extra verification.
4	Average Loan by Age Groups	Average LoanAmount per group: Adults 127,901; Middle aged 127,459; Senior Citizens 127,355; Teen 126,674.	Loan size barely changes with age (under 1%), so exposure per loan is similar across life stages.
5	Default Rate by Year	Defaulted loans divided by loans issued in each year: 11.6%, 11.5%, 11.7%, 11.7%, 11.5%, 11.6% for 2013 to 2018.	A flat trend within 0.2 points. This is the baseline for monitoring.

Key takeaway. Default risk is steady over time (11.5% to 11.7% a year) but differs by borrower type: unemployed borrowers default at 13.6% against 9.5% for full-time borrowers, while loan size and income barely differ between groups.

Dashboard Page 2: Applicant Demographics and Financial Profile

This page answers the question: who is borrowing, and how is lending distributed?



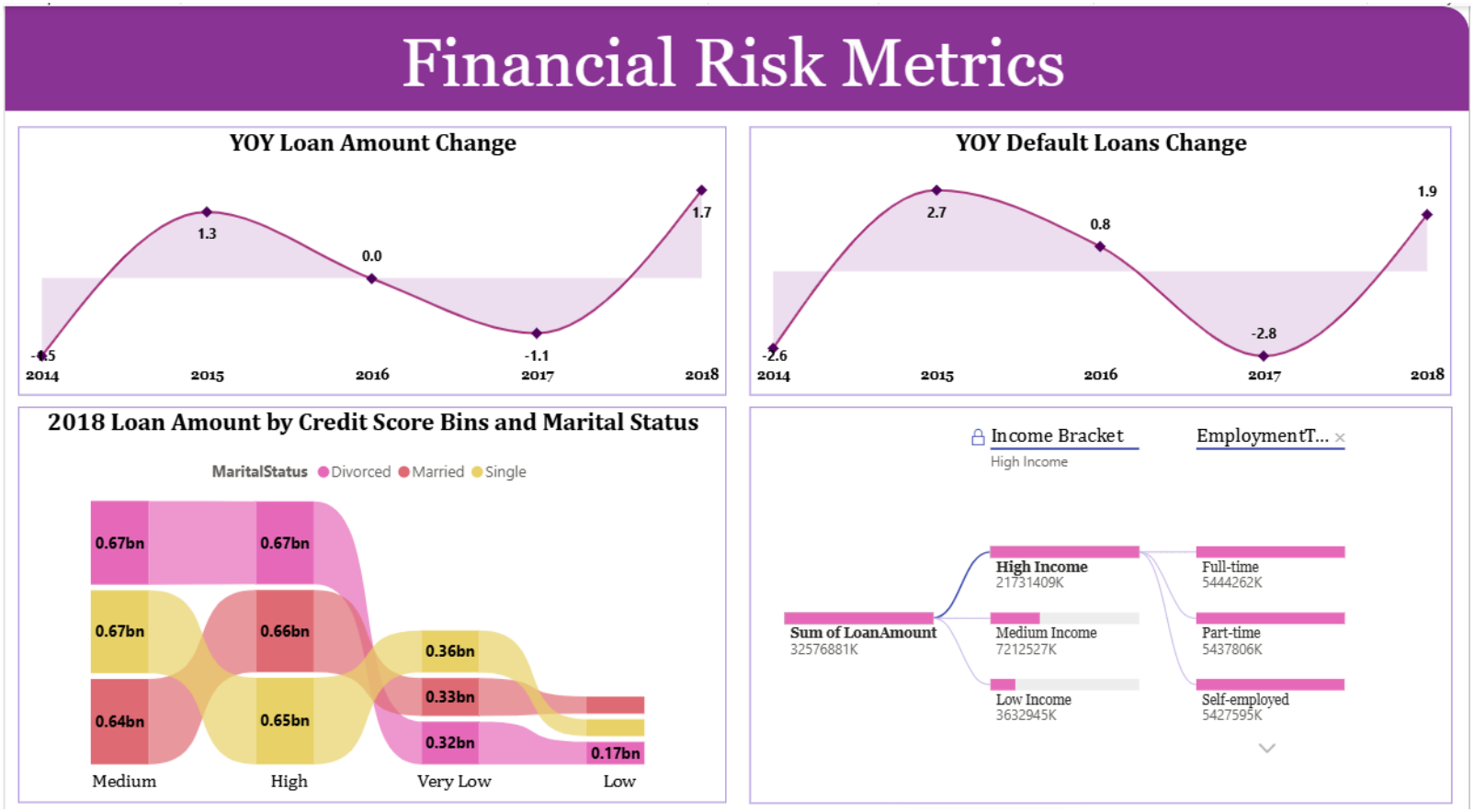
Page 2: Visual Details

#	Visual	What it shows and how it is calculated	How to read it and use it
6	Median by Credit Score Bins	Median LoanAmount per credit bin: Low 128,397; Medium 127,764; Very Low 127,515; High 127,149.	The typical loan is the same in every credit tier (within 1%): credit quality does not change how much is lent.
7	Average Loan (High Credit) by Age Group and Marital Status	Average LoanAmount for the High credit bin across 12 age-group and marital-status combinations (123.97K to 128.57K).	Read the K values. The percentages are each slice's share of the combined averages (about 8.3% each). Loan sizes for creditworthy borrowers are consistent across segments.
8	Total Loan (Adults) by Credit Bins	Sum of LoanAmount for Adults (20 to 39) per credit bin: Medium 4.6bn, High 4.5bn, Very Low 2.3bn, Low 1.1bn.	About 73% of adult lending sits in the Medium and High bins. Totals follow the number of loans per bin; the Very Low and Low exposure (3.4bn) is the segment to watch.
9	Total Loan (Middle Aged) by Mortgage and Dependents	Sum of LoanAmount for ages 40 to 59 by mortgage and dependents status: about 3.1bn in each of the four combinations.	Exposure is evenly split, so there is no concentration by household type.
10	Number of Loans by Education Type	Count of loans per education level: Bachelor's 64,366; High School 63,903; Master's 63,541; PhD 63,537.	The mix is balanced (about 25% each). Default rate does differ by education: High School 12.9%, PhD 10.6%.

Key takeaway. Loan sizes are nearly identical across credit tiers, age groups and household profiles, so differences between borrower groups show up in default rates rather than in how much is lent.

Dashboard Page 3: Financial Risk Metrics

This page answers the question: how are lending volume and defaults moving over time?



Page 3: Visual Details

#	Visual	What it shows and how it is calculated	How to read it and use it
11	YoY Loan Amount Change	(Loan amount this year - last year) / last year, in percent: 2014 -1.5; 2015 +1.3; 2016 0.0; 2017 -1.1; 2018 +1.7.	Lending volume is stable, within plus or minus 2%. 2013 has no prior year and is not shown.
12	YoY Default Loans Change	Percentage change in the number of defaulted loans: 2014 -2.6; 2015 +2.7; 2016 +0.8; 2017 -2.8; 2018 +1.9.	Defaults move slightly more than lending volume, in line with the small rise in default rate in 2015 to 2016 and the fall in 2017. Read with visual 11.
13	2018 Loan Amount by Credit Score Bins and Marital Status	Sum of LoanAmount from 1 January to the latest loan date (31 December 2018), by credit bin and marital status.	Shows latest-year lending, about one-sixth of the 32.6bn six-year total. Marital statuses carry similar amounts within each bin.
14	Loan Amount by Income Bracket and Employment Type	Decomposition tree: total 32.58bn split by income bracket (High 21.73bn, Medium 7.21bn, Low 3.63bn), then by employment type (about 5.4bn each within High).	An interactive drill-down. High income holds 66.7% of the amount, matching its 66.7% share of loans, because that bracket spans the widest income range.

Key takeaway. Lending volume and defaults move together within a narrow band (under 3% a year), and income-bracket totals follow how wide each bracket is rather than any difference in lending behaviour.

5. Key Insights

The dashboard's default-rate views were extended by calculating default rate for every borrower and loan characteristic in the data. Each breakdown reconciles to 255,347 loans and 29,653 defaults. Figure 2 ranks the variables by the gap between their highest- and lowest-risk groups.

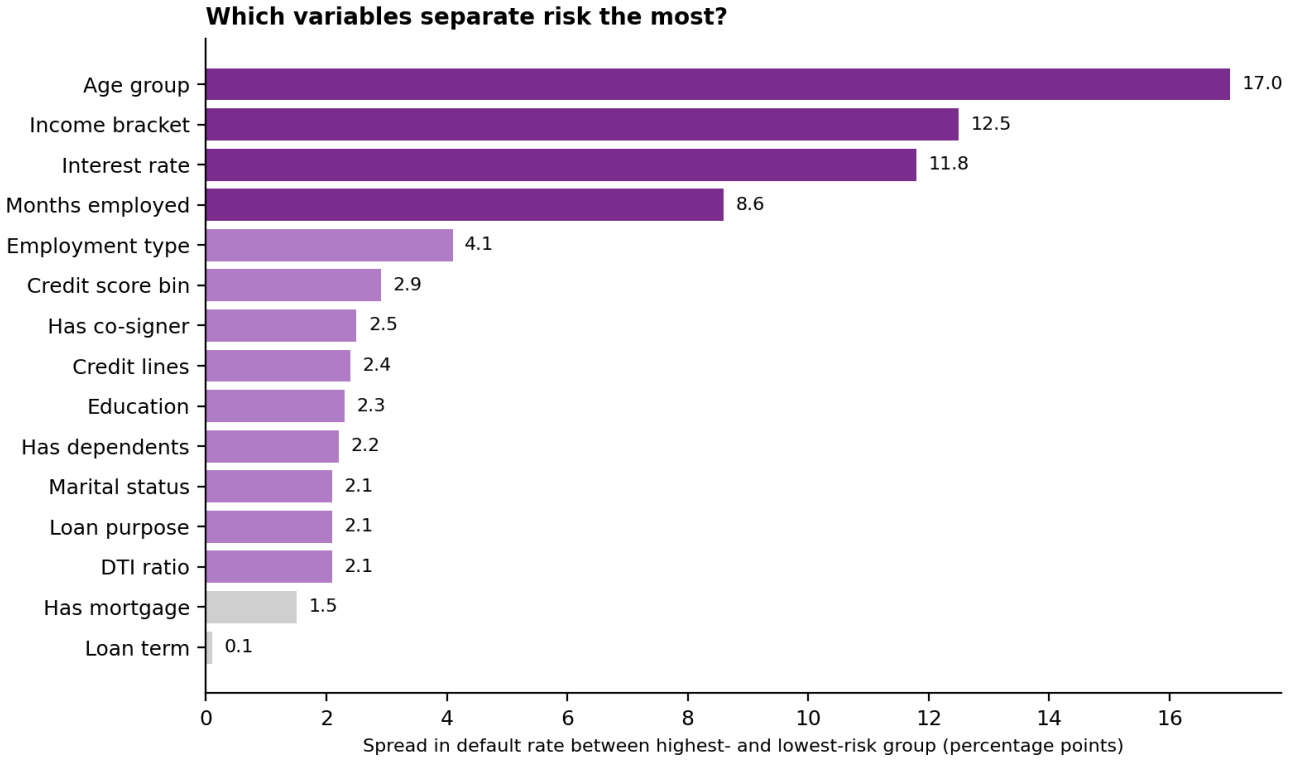
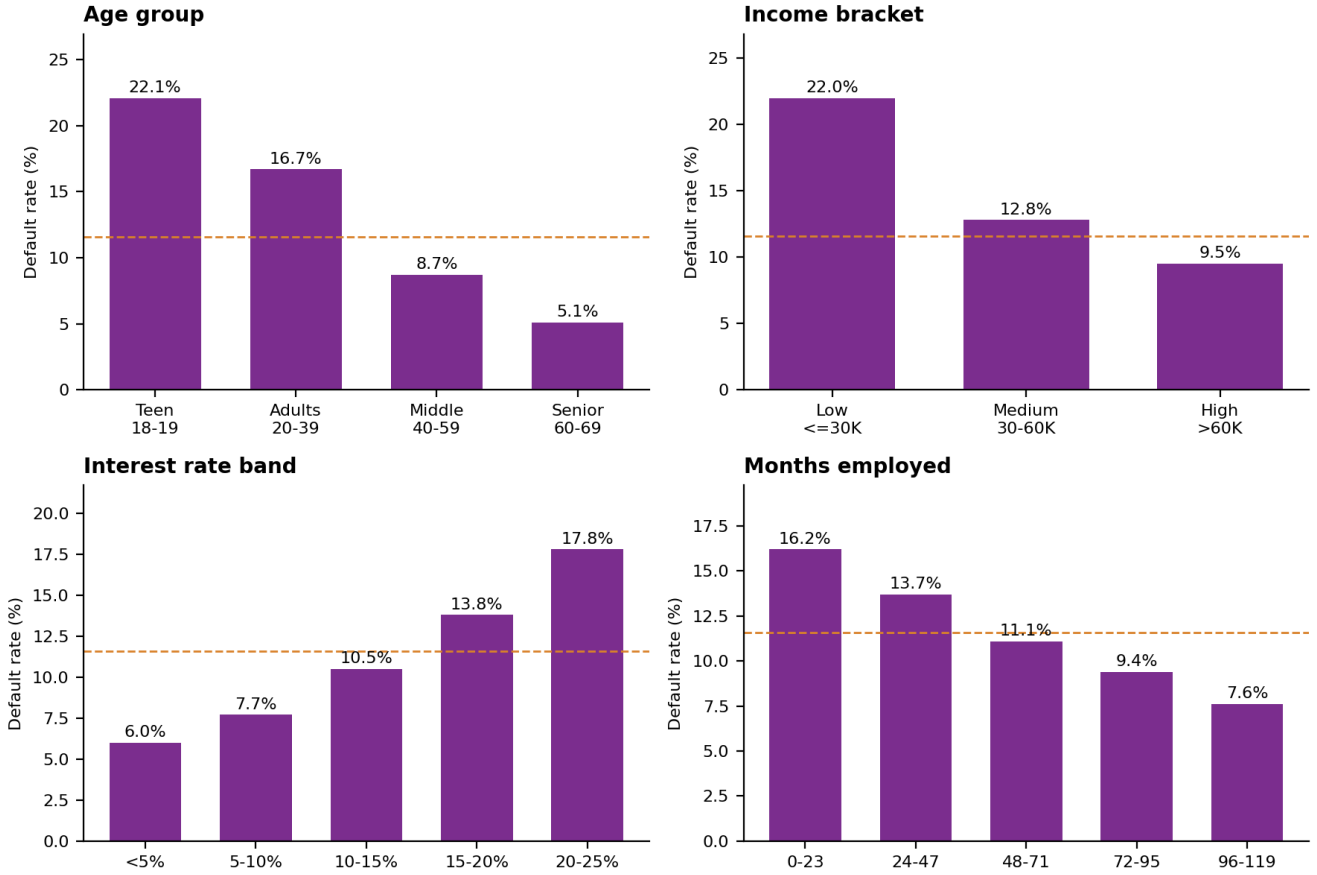


Figure 2. Gap in default rate between the highest- and lowest-risk group of each variable (percentage points).

How to read the insights. Groups are large (at least 9,800 loans, usually 28,000 to 128,000), so differences of more than about 1 point are real patterns in this dataset. Variables overlap, for example age with tenure and income, and no adjustment was made for that. Every finding is therefore an association, not a cause.

5.1 The Four Strongest Risk Separators

Age, income, interest rate and job tenure separate low-risk from high-risk loans by roughly 9 to 17 percentage points.



Dashed line = overall default rate (11.6%). Each bar group contains 28,000+ loans (10K+ for Teen).

Figure 3. Default rate by group. Dashed line: overall default rate 11.6%.

Insight 1: The portfolio default rate is stable

Finding	About 1 in 9 loans defaults, and the rate has been steady since 2013.
Evidence	29,653 defaults in 255,347 loans (11.6%). Yearly rates stay between 11.5% and 11.7%. Lending volume and defaulted loans each change by less than 3% a year.
Interpretation	No lasting deterioration or improvement. Year-to-year movement is small next to the gaps between customer segments.
Business implication	Risk depends mainly on who borrows, so segment monitoring is more informative than the overall rate.
Recommendation	Use 11.6% as the baseline with alert bands, and track segment rates beside it.
Limitation	Six annual points and no economic context.

Insight 2: Younger and lower-income borrowers default far more, yet receive the same loan size

Finding	Default is concentrated in young and low-income borrowers, and loan size does not reflect that.
Evidence	Age: Teen (18 to 19) 22.1%, Adults (20 to 39) 16.7%, Middle aged (40 to 59) 8.7%, Senior (60 to 69) 5.1%. Income: Low (up to 30K) 22.0%, Medium 12.8%, High (above 60K) 9.5%. Average loan is about 127K to 128K in every group; for the low-income bracket that is at least 4.3 times annual income (derived).
Interpretation	Loan size does not follow risk. Age and income probably stand in for stability and affordability; this was not tested.
Business implication	The riskiest groups carry the same exposure per loan as the safest.
Recommendation	Add default-rate visuals by age and income, and test a loan-to-income measure and limits for the high-risk segments.
Limitation	Association only. Age is a sensitive characteristic in credit decisions and needs legal and fairness review.

Insight 3: Interest rate and job tenure are strong risk signals

Finding	Higher interest rates and shorter employment go with much higher default.
Evidence	Interest rate band: below 5% 6.0%; 5 to 10% 7.7%; 10 to 15% 10.5%; 15 to 20% 13.8%; 20 to 25% 17.8% (the 25%+ band has only 53 loans and is excluded). Months employed: 0 to 23 16.2%; 24 to 47 13.7%; 48 to 71 11.1%; 72 to 95 9.4%; 96 to 119 7.6%.
Interpretation	Rates likely reflect the lender's risk view and the repayment burden. Tenure separates risk about twice as strongly as employment type (8.6 vs 4.1 points).
Business implication	Both are practical early-warning signals, and neither is on the current dashboard.
Recommendation	Add interest rate and tenure bands as dashboard fields, and track the share of loans priced above 15%.
Limitation	High rates may reflect loans already seen as risky (reverse causality). Tenure overlaps with age.

Insight 4: Employment type, co-signer, dependents and mortgage add smaller, consistent differences

Finding	Several profile factors separate risk by about 1.5 to 4 points.
Evidence	Employment: Unemployed 13.6%, Full-time 9.5%. Co-signer: 10.4% with, 12.9% without. Dependents: 10.5% with, 12.7% without. Mortgage: 10.9% with, 12.3% without. Education: High School 12.9%, PhD 10.6%. Marital status: Divorced 12.5%, Single 11.9%, Married 10.4%.
Interpretation	A co-signer adds a second person responsible for repayment, a plausible protective effect. The other factors likely reflect stability.
Business implication	A co-signer requirement is a lever the lender controls.
Recommendation	Test co-signer requirements for higher-risk segments and check the gap holds within each segment.
Limitation	Co-signers may be added mainly to safer loans.

Insight 5: Credit score, DTI and loan term separate risk weakly in this data

Finding	Variables often relied on show modest or no separation.
Evidence	Credit score bins: 13.3% (Very Low) to 10.4% (High). DTI ratio: about 10.4% to 12.4%. Loan term: 11.6% to 11.7% for all five terms. Loan purpose: Home 10.2% against 11.8% to 12.3% for the rest.
Interpretation	In this dataset they carry less information than age, income, interest rate and tenure.
Business implication	A scorecard based only on credit score would miss most of the visible risk differences.
Recommendation	Review the credit bin cut-offs and test finer bins; treat Home loans as a lower-risk product in monitoring.
Limitation	Bins are unequal in width and results depend on them; other portfolios may differ.

Insight 6: Defaults hold a larger share of the money than of the loan count

Finding	Money at risk is higher than the headline default rate suggests.
Evidence	4.29bn of 32.58bn lent defaulted (13.2%), against 11.6% by count. The implied average defaulted loan is about 145K against 127.6K overall (derived). Share of amount by purpose: Business 13.9%, Auto 13.5%, Other 13.4%, Education 13.3%, Home 11.6%.
Interpretation	Larger loans are somewhat more likely to default, so count-based rates understate exposure.
Business implication	Exposure and potential loss are higher than the count-based rate shows.
Recommendation	Report count-based and amount-weighted default rates side by side, and test default rate by loan-size band.
Limitation	Defaulted amount is not loss (no recovery data). Derived values use rounded totals.

6. DAX Measures and Calculations

Filter context is the key idea: each chart point (for example one employment type) filters the table before a measure runs, so one formula returns a different value at every point. **CALCULATE** changes the filter context, **ALLEXCEPT** removes all filters except the columns named, and **DIVIDE** divides safely when the denominator may be zero.

Measure	What it calculates
loan default by purpose	Sum of LoanAmount where Default = True.
Average Income by employment type	Average Income, keeping only the employment-type filter.
Default rate by employment type	Defaulted loans divided by all loans in the same employment type.
Average Loan by age groups	Average LoanAmount per age group.
Default Rate by Year	Defaulted loans divided by all loans in the same year.
Median by Credit score	Median LoanAmount over the loans in each credit bin.
Average loan amt(High credit)	Average LoanAmount over loans in the High credit bin.
Total loan(adults) by credit bins	Sum of LoanAmount where age group is Adults.
Total loan Amount (middle aged)	Sum of LoanAmount where age group is Middle aged.
loan by education type	Count of loans (rows with a LoanID).
YOY Loan amount change	(Current year amount - prior year amount) / prior year amount x 100; blank for the first year.
YOY default loans change	The same calculation on the number of defaulted loans.
YTD loan Amount	Sum of LoanAmount from 1 January to the latest date in view, keeping only the credit bin and marital status filters.

6.2 Key Measures and Validation

```
loan default by purpose =
CALCULATE(SUM(Loan_default[LoanAmount]), Loan_default[Default] = TRUE())

Default rate by employment type =
DIVIDE(
    CALCULATE(COUNTROWS(Loan_default), Loan_default[Default] = TRUE()),
    COUNTROWS(Loan_default)
)

YOY Loan amount change =
VAR CurrentYear = YEAR(MAX(Loan_default[Loan_Date_DD_MM_YYYY]))
VAR CurrentAmount =
    CALCULATE(SUM(Loan_default[LoanAmount]), Loan_default[Year] = CurrentYear)
VAR PriorAmount =
    CALCULATE(SUM(Loan_default[LoanAmount]), Loan_default[Year] = CurrentYear - 1)
RETURN DIVIDE(CurrentAmount - PriorAmount, PriorAmount) * 100
```

In the default rate measure, the top of DIVIDE counts defaulted loans in the current group and the bottom counts all loans in the same group. In the YoY measure, each VAR is calculated once, then DIVIDE returns blank when there is no prior year.

Default Rate by Year uses ALLEXCEPT so that each year is compared with its own loans.

```
Default Rate by Year =
VAR TotalLoans = CALCULATE(COUNTROWS('Loan_default'),
                           ALLEXCEPT('Loan_default','Loan_default'[Year]))
VAR DefaultLoans = CALCULATE(COUNTROWS('Loan_default'),
                             'Loan_default'[Default] = TRUE(),
                             ALLEXCEPT('Loan_default','Loan_default'[Year]))
RETURN DIVIDE(DefaultLoans, TotalLoans, 0)
```

Validation. The measures were tested against queries on the data and refined in four places: the defaulted amount by purpose now includes the Default condition; the employment default rate now divides by each group's own loans; and both YoY measures now leave the first year blank. The education chart is titled as a loan count and the YTD chart as the 2018 amount, so each title matches its calculation.

7. Recommendations and Future Scope

Strengthen monitoring (descriptive)

- Add a risk-drivers page showing default rate by age group, income bracket, interest rate band and months employed, with year, purpose and employment slicers.
- Show count-based and amount-weighted default rates side by side, and add a loan-to-income measure.
- Set alert bands around the 11.6% baseline and for high-risk segments (for example under-40 borrowers, low income, under two years of tenure, rates above 15%).
- Sort credit score bins and age groups in risk order, and add a date table with a Year slicer so the YTD view can cover any year.

Deeper analysis

- Cross-tabulate age, income and tenure to see how much of each effect is shared.
- Test the gaps with confidence intervals or chi-square tests, and check that segment gaps hold in every year.

Predictive modelling (future, not part of this project)

The current work is descriptive: it shows what happened and where. A predictive model, which estimates the risk of a new loan, would be a separate step: start with logistic regression, then tree-based models, evaluated on unseen data with ROC-AUC and calibration, with care over class imbalance (about 12% defaults), fairness and regulatory review of sensitive variables. Age, income, interest rate, tenure, co-signer and number of credit lines are the first candidates.

Engineering improvements

- Move the age, credit and income bin definitions into a SQL view or the Dataflow so other reports can reuse them.
- Record refresh schedules and failure alerts for both the Dataflow and the dataset, and review gateway availability.

8. Limitations

Data

- **Dataset origin:** the source is not recorded, and the evenly balanced groups suggest a public or simulated dataset. Findings describe this dataset and may not hold in a real portfolio.
- **Definitions:** how Default is defined, the observation period per loan, and how income and tenure were verified are not documented.
- **Approved loans only:** rejected applicants are absent, so approval policy cannot be assessed.

Method

- **Association, not causation:** variables overlap (for example age with tenure and income) and no multivariate adjustment was made.
- **Unequal bins:** credit and income bins differ in width, so bin totals reflect group size more than behaviour; rates are the more reliable comparison.
- **Derived figures:** amount-weighted rates, the average defaulted loan and loan-to-income are calculated from rounded chart totals and are approximate. Defaulted amount is not loss.

Technical

- **Not reviewed:** gateway configuration, credentials, refresh schedules and the SQL load process.
- **Chart axes:** several charts use zoomed axes, which makes differences under 1% look large; read the labels as numbers.

Before using these findings for lending decisions

- Confirm the dataset's source and definitions, and validate the findings on the lender's own data.
- Build and test multivariate models on unseen data to separate overlapping effects.
- Complete a fairness, legal and regulatory review of any use of age, marital status or similar characteristics.